
HOMEWORK 1 (MODERN OPTIMIZATION ALGORITHMS, MATH 485)

Due date: See class webpage on Canvas.

1. Let $A \subset \mathbb{R}$ and $B \subset \mathbb{R}$ be nonempty sets. Prove that if

$$(\forall a \in A)(\forall b \in B) \quad a \leq b \tag{1}$$

then $\inf A \leq \inf B$ and $\sup A \leq \sup B$.

2. Let $A \subset B \subset \mathbb{R}$ be nonempty sets and suppose that $A \subset B$. Show that $\sup A \leq \sup B$ and $\inf A \geq \inf B$.
3. Let $f: \mathcal{H} \rightarrow [-\infty, +\infty]$ and let $C \subset \mathcal{H}$. Then **image of C under f** is defined by $f(C) = \{f(c) \mid c \in C\}$. Suppose that $g: \mathcal{H} \rightarrow [-\infty, +\infty]$ satisfies

$$(\forall c \in C) \quad g(c) \leq f(c). \tag{2}$$

- (a) Show that $\inf g(C) \leq \inf f(C)$ and $\sup g(C) \leq \sup f(C)$.

EC: Why can't we set $A = g(C)$, $B = f(C)$, and use Problem 1?

- (b) Show that, if $\text{dom } f \cap \text{dom } g \neq \emptyset$, then $\text{dom}(f + g) \neq \emptyset$.

4. Let $p \in]0, +\infty[$, let $C \subset \mathcal{H}$, and let $f: \mathcal{H} \rightarrow [-\infty, +\infty]$ be a nonnegative function, i.e.,
 $(\forall x \in \mathcal{H}) \quad f(x) \geq 0$.

- (a) Show that exponentiation by p preserves order on $\mathbb{R}_+$, i.e.,

$$(\forall \alpha_1, \alpha_2 \geq 0) \quad \left[\alpha_1 \leq \alpha_2 \quad \Rightarrow \quad \alpha_1^p \leq \alpha_2^p \right]. \tag{3}$$

- (b) Prove that

$$\operatorname{argmin}_{x \in C} f(x) = \operatorname{argmin}_{x \in C} (f(x))^p$$

(hint: this requires nonnegativity of f ; use (4(a)).)

- (c) Assuming 4(b) is true, does this mean that, for every $p \in]0, +\infty[$, $\inf f(C) = \inf f^p(C)$? Prove this if true; otherwise, provide a counterexample.